Software Development Project 1

Welcome to the Software Development Project 1 course! 👋

Introduction to software development projects

- The learning objectives for this week are:
  - Knowing about the course contents, assessment and schedule
  - Knowing how software development is carried out in practice and what kind of common problem areas it has
  - Knowing what is agile software development
  - Knowing what is the software development lifecycle and how it fits the agile software development
  - Knowing what is Scrum and how it is related to the agile software development
  - Knowing how stakeholders’ requirements are managed in agile software development

About the course

- So far during the studies we have mostly worked on fairly small applications by ourselves
- Different kinds of problems arise while we work on more complicated applications in a software development team
- These problems aren’t only technical, the management of the software development process can be quite tricky
- During this course, we will learn how to manage the software development process and how to work as a member of a software development team
- We will learn these skills in practice by developing a software development project in a software development team consisting of 4-5 students

About the course

- Each team will be developing a quiz management application described in more detail on the course page
- The final application will be a full stack web application in which a frontend communicates with a backend
- The implementation technologies of the application are Spring Boot and React
- The development of the project is split into three two-week iterations called Sprints
- Each Sprint has high-level requirements, but teams should make most of the technical decisions themselves
- The project starts on week three

Assessment

- The assessment is based on the **team's project** and the **personal contributions** of a team member
- The project assessment is based on the following aspects:
  - Technical implementation
  - Project management
  - Documentation
  - Following the schedule
- Based on these assessment criteria, the team receives a **project grade** (0-5)

Assessment

- Each team member’s personal assessment is based on the following aspects:
  - Activity in teamwork
  - Technical contributions
  - Project management and documentation contributions
  - Exercise submissions
- The personal assessment is done based on the teacher’s observations and peer reviews from the team members
- Each team member receives a **personal grade** (0-5) based on their personal efforts towards the project and individual exercises

Assessment

- The final grade (0-5) is composed of the project grade and the personal grade
- The following are necessary to pass the course:
  - At least 16 of the first two week’s exercises have to be completed to pass the course
  - Written peer review for each team member
  - Passing personal grade
  - Written final report

Schedule

- There are weekly sessions during which we will cover different topics
- Attendance on weekly sessions is **mandatory**
- During the first two weeks, we will cover topics that are important to grasp before starting to work on the project
- The teams will start working on the project during the second week
- During the teamwork we will learn about new topics, but most importantly we will learn how to apply our knowledge in practice
- Both individual exercises and project exercises have **deadlines**
- The detailed schedule can be found on the course page

What makes a software development project?

- So far during the studies we have implemented software mostly alone with more or less detailed instructions on what to do
- In the real world software is developed in **software development projects** by a diverse group of people referred to as the **software development team**
- Depending on the budget and scope of the project, the timespan of a software development can be anything from a few months to many years

Stakeholders

- A software development project affects many people in different ways
- For example, the software's end users get to use features they desire and the customer gets financial value for the project they are funding
- The people or groups of people affected by a software development project are referred to as the software development project's **stakeholders**
- On top of technical skills, the future of software developer's work will most likely consist more and more of understanding the needs of the stakeholders and applying technical solutions to meet those needs

Software development team

- A software development team is responsible for the outcome of the project
- It is a diverse and self-organized group of people consisting of not only technical people
- The team can consist for example of a representative of the end users (commonly referred to as a product owner), user-experience (UI) and user-interface (UX) designers and software developers
- The size of the team varies, but they commonly consist of fewer than 10 people
- So that a software development team can work together efficiently they need to follow a **common process** in their day-to-day work
- This process is referred to as a **software development process**

Problems with software development projects

- Developing software based on the stakeholders’ needs in a software development team isn’t simple and many things can (and often do) go wrong
- Which problems do you think software development projects commonly have? Are the problems technical such as programming errors or something else?

Problems with software development projects

- Based on a study conducted in 2022, **70% of software development problems are non-technical**. The three most common problems raised were:
  - **Management issues**: excessive product owners, unclear task descriptions, unrealistic schedules, and constantly changing requirements
  - **Miscommunication and lack of communication**: the development team interprets project requirements differently from what the client had in their mind
  - **Customer-related issues**: customers lack understanding of technical processes, leading to unrealistic expectations and tight deadlines that developers struggle to meet

Agile principles in software development

> *"able to move quickly and easily."* — Dictionary definition for the word *"agile"*

- The word *"agile"* is often used in many industries to describe the way of working in organizations
- The word is commonly used in a positive manner, for example, *"we are an agile organization"*, or *"we work in an agile manner"*
- In **agile software development**, the development process follows values and principles that have been found to lead to successful software development projects
- These values and principles have been constituted and written down as the **Manifesto for Agile Software Development** by famous software development pioneers

Manifesto for Agile Software Development

- The Manifesto for Agile Software Development describes the following values:
  - **Individuals and interactions over processes and tools**: people should communicate directly and work together effectively instead of relying too much on rigid processes or tools
  - **Working software over comprehensive documentation**: the main focus should be on delivering functioning software, while keeping documentation useful but not excessive.
  - **Customer collaboration over contract negotiation**: the team should work closely with the customer throughout the project instead of only following an initial agreement
  - **Responding to change over following a plan**: plans are important, but the team should be ready to adjust them when requirements or circumstances change

Agile principles in software development

> *"Responding to change over following a plan."*

- One of the key values is the attitude towards **change**
- For example, there is often a need to change the software's requirements during the development process
- Agile software development process should welcome any kind of change with open arms
- That is, because change is inevitable and frequent in many business environments
- The Manifesto for Agile Software Development doesn't go into detail on how to actually implement these values in practice
- Different Agile software development process frameworks, such as **Scrum** and **SAFe** describe a detailed process that follows these values

Software development lifecycle

- It requires **different phases** to be completed so that an idea of what we can do with a software becomes an actual working software which provides the desired features
- The software development process is divided into different phases:
  1. Requirements phase
  2. Design phase
  3. Implementation phase
  4. Test phase
  5. Deployment phase
  6. Maintenance phase
- The phases are commonly performed in the mentioned order and the whole process is often called the **software development lifecycle**

Requirements phase

- In the **requirements phase**, the development team collects requirements from several stakeholders such as customers and end users
- Requirements describe the features and behaviors of a system from the user's perspective, focusing on **how users interact with it and the outcomes they want to achieve**
- For example, *"As a blog reader I want to browse list of blog posts of a blog so that I can find interesting posts to read"* could be a requirement for a blog application
- These requirements are commonly documented and managed in so-called *"backlogs"* which act as a prioritized list of work for the development team

Design phase

- In the **design phase**, the development team analyzes requirements and identifies the best solutions to create the software
- For example, they may consider integrating pre-existing modules, making technology choices, and identifying development tools
- During the design phase different kinds of documentation, such as architecture diagrams, are produced to support the **implementation phase**

Implementation phase

- In the **implementation phase**, the development team codes the product
- They analyze the requirements to identify smaller coding tasks they can do daily to achieve the final result
- The organization of the collaboration during the implementation phase requires the development team to carefully follow mutually agreed **process**
- The development team needs to use different kinds of tools to ease the collaboration, such as **version control tools**, which we will cover later

Test phase

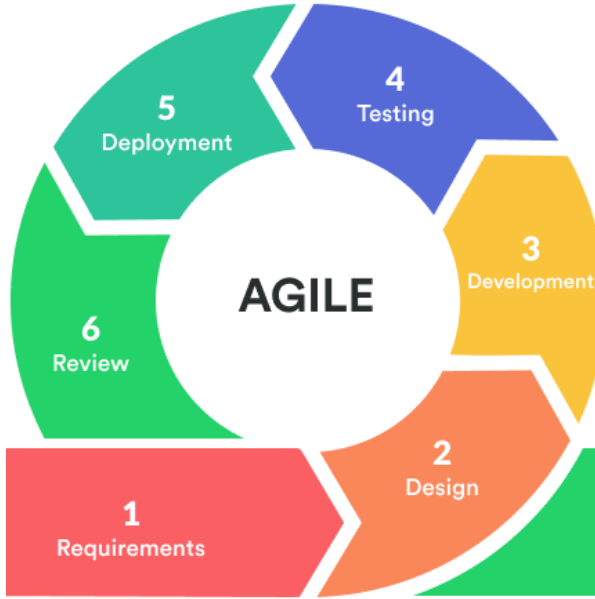
- In the **test phase**, the development team combines automation and manual testing to check that the software works as intended
- In practice, the test phase isn’t usually separated from the implementation phase
- This means that software developers usually implement a small coding task, write automated test cases for the task and move on to the next task
- This practice is even taken so far that the tests are written **before** the implementation which is referred to as test-driven development (TDD)

Deployment and maintenance phase

- In the **deployment phase**, the implemented software is distributed to the users
- For example, a web application is published online so that users can access it with their browsers
- Once the software is distributed, it needs to be constantly **maintained**
- In the **maintenance phase**, among other tasks, the development team fixes bugs, resolves user issues, and manages software changes

Agile software development lifecycle

- Agile software development has an **iterative approach** in the software development lifecycle
- In this approach, the software is developed in short, typically one or two-week-long iterations
- Each iteration starts with the requirements phase and during the iteration design, implementation, test, deployment, and maintenance phases are completed
- The outcome of each iteration is working software that users can actually use and give feedback
- This cycle repeats in every iteration



Benefits of the iterative approach

> *"When to use iterative development? You should use iterative development only on projects that you want to succeed."*
> —Martin Fowler, author and international public speaker on software development

- The benefits of the iterative approach are the ability to respond to change quickly and the feedback loop it provides
- After each iteration the requirements can change which makes it easy to respond to new user and business needs
- After each iteration the users of the software can get their hands on new features which they can give feedback on
- The feedback can be used to define requirements for the next iterations
- This forms the feedback loop which is the heartbeat of the agile software development process

Scrum

- **Scrum** is a widely adopted iterative software development process framework that defines practical ways to carry out agile software development principles
- Based on the State of Agile report of the year 2022, 87% of respondents used Scrum at the team level
- In the Scrum process, the **Scrum Team** develops software in fixed-length iterations called **Sprints**
- Sprints should be kept short, preferably at most four weeks long
- In each Sprint, there are fixed **events**, which help the Scrum Team organize their work and keep track of the progress of the Sprint
- During the course we will use Scrum to manage our software development process while working on the project

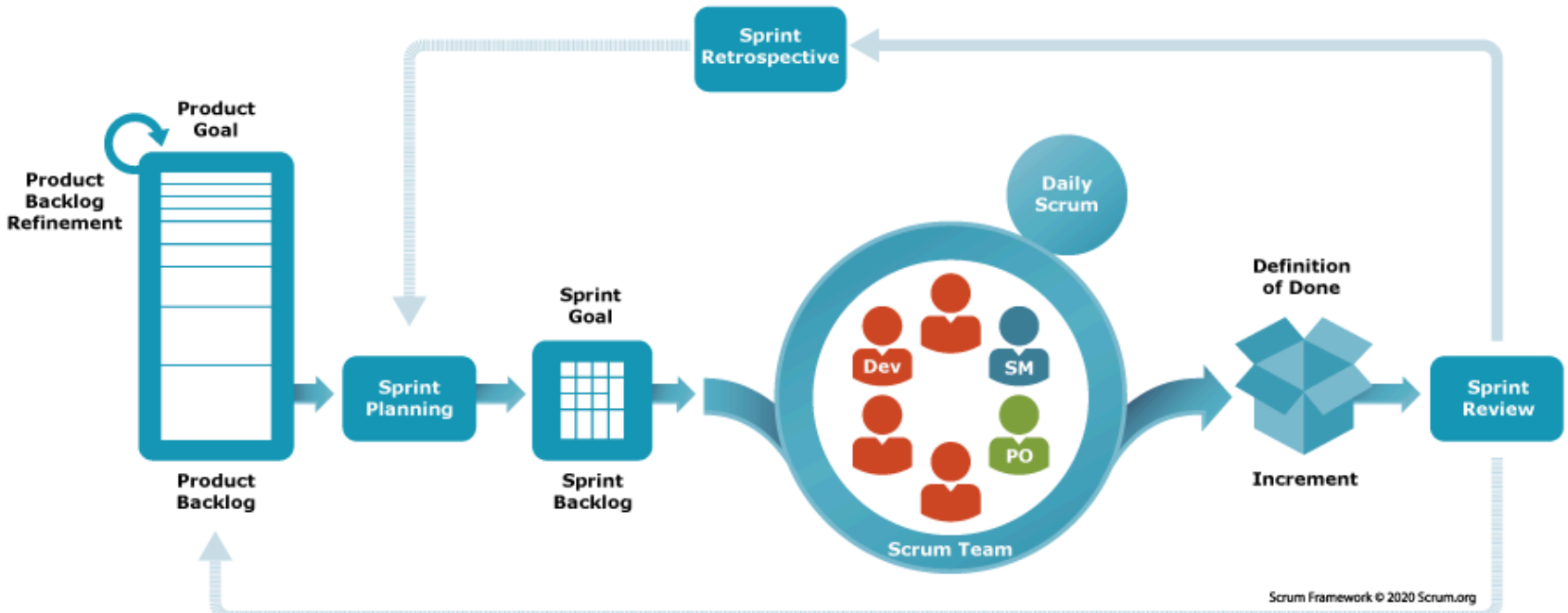
Scrum Team

- The **Scrum Team** is responsible for the outcome of each Sprint
- It consists of one **Scrum Master**, one **Product Owner**, and **Developers**
- To avoid coordination difficulties, the size of the Scrum Team should be fairly small, preferably at most 10 people
- **Developers** are responsible for the technical implementation of the software
- **Product Owner** is responsible for maximizing the value of the product resulting from the work of the Scrum Team
- **Scrum Master** is responsible for ensuring that the software development process follows the Scrum practices
- During the course, the teacher is the Product Owner and students are the Developers

Scrum Events

- Four *events* take place during each Sprint: **Sprint planning**, **Daily Scrum**, **Sprint Review** and **Retrospective**
- At the beginning of the Sprint, there's a **Sprint planning** event during which the Scrum Team decides which set of requirements are implemented during the upcoming Sprint
- During the Sprint, the Scrum Team organizes **Daily Scrum** meetings to discuss the progress of the Sprint
- At the end of the Sprint, there's a **Sprint Review** event during which the Scrum Team goes through the outcome of the Sprint
- Before starting the next Sprint, the Scrum Team discusses the problems with the process and figures out ways to improve it during the **Retrospective** event

The Scrum process



Requirements

- During the requirements phase, the stakeholders' requirements for the software are defined and documented
- To derive the requirements, the development team needs to have a clear and thorough understanding of the software under development and continuously communicate with the stakeholders
- Software's requirements are commonly categorized as **functional** and **non-functional requirements**
- Functional requirements are the requirements stated by the user which one can see directly in the final product
- Non-functional requirements on the other are quality constraints related to e.g. performance and security that the system must satisfy

Functional and non-functional requirements

- For example the following requirements are functional requirements:
  - "User should be able to register with a username and password"
  - "User should be able to see a list of latest blog posts of their favorite blogs"
- And the following requirements are non-functional requirements:
  - "User’s password should be stored in the database as a Bcrypt password hash upon registration"
  - "List of latest blog posts of user’s favorite blogs should be loaded on average in a less than one second"

Agile requirement specification

- In agile software development, requirements are commonly written as **user stories**
- A user story is a short, simple description of a feature told from the perspective of the person who desires the new feature in the software
- During each Sprint, the Developers of the Scrum Team implement features for the software based on user stories
- User stories are sort of to-do items, which help the Scrum Team understand and remember what kind of features need to be implemented for the software
- **Use cases** are another, less Agile-friendly, way of defining requirements. They focus more on **how the system will work**, not **what the user needs**

User story

- For example, two separate user stories for a blog application could be the following:
  - *"As a content **creator**, I want to **create a new blog** so that **I can start writing blog posts**"*
  - *"As a blog **reader**, I want to **browse list of blog posts of a blog** so that **I can find interesting posts to read**"*
- A user story is written from the user's perspective and commonly follows the following format:

> As **[a user persona]**, I want **[to perform this action]** so that **[I can accomplish this goal]**.

Let’s write some user stories

- Come up with and write down user stories in one or more applications you have used, for example Netflix or Moodle
- Use the *"As [a user persona], I want [to perform this action] so that [I can accomplish this goal]"* format
- Try to come up with different user personas for the user stories
- Share your user stories with others and let’s discuss about them

Writing good user stories

- A good user story describes a **feature that provides value for the end user of the software**
- This means, that the description should be written from the user's perspective and not contain technical details
- For example the following user story is described from the developer's point of view:

> ❌ As a developer, I want to optimize the database by adding a new index to the blog posts table so that the list of blog posts of a blog loads fast.

- A better user story would express the feature from the user's point of view:

> ✅ As a blog reader, I want the list of blog posts of a blog load fast so that I can find interesting blog posts to read quickly.

The INVEST criteria

- The **INVEST criteria** is one popular guideline for a good user story:
  - **Independent**: written so they can be developed and tested independently of other stories
  - **Negotiable**: written to allow for negotiation between the development team and the customer
  - **Valuable**: should provide value to the end user.
  - **Estimable**: written in a way that allows the development team to estimate the amount of effort required to complete them
  - **Small**: small enough to be completed within a single iteration of the development process
  - **Testable**: written to allow testing to be performed at the end of the development process

The INVEST criteria

- Which INVEST criteria (independent, negotiable, valuable, small, testable) does the following user story violate?

> ❌ As a blog reader, I want to be able to search blog posts by name, category and author so that I can find relevant blog posts to read

The INVEST criteria

- It is better to split these kinds of big user stories into multiple smaller user stories:

> ✅ As a blog reader, I want to be able to search blog posts by name so that I can find relevant blog posts to read
>
> ✅ As a blog reader, I want to be able to search blog posts by category so that I can find relevant blog posts to read
>
> ✅ As a blog reader, I want to be able to search blog posts by author so that I can find relevant blog posts to read

- This allows faster delivery and feedback and lower risks of requirement misunderstandings and technical issues

- If the description of the feature contains words like *"and"* or *"or"*, it could be that the user story can be split into smaller user stories